

NBA Player Trajectory Modeling: Predicting Peak Performance Windows and Championship Contention

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Abstract

Identifying when an NBA player is performing at their peak and projecting which teams are best positioned for a championship run are central challenges in modern sports analytics. We present a two-stage data-driven framework applied to 27 seasons (1996–2023) of per-season statistics for 2,551 players. In the first stage, we construct a composite performance score from z-score normalized efficiency metrics and apply K-Means clustering to classify players into four career trajectory archetypes: *Short-Career Burners*, *Late-Peak Veterans*, *Rising Stars*, and *Elite Long-Tenured Stars*. In the second stage, we train three supervised classifiers—Logistic Regression, Random Forest, and XGBoost—to predict whether a given player-season represents that player’s peak performance window, achieving an AUC of 0.8348 with XGBoost. Composite performance score, net rating, and true shooting percentage are the dominant predictors of peak seasons. Applied to 2022–23 rosters, our model correctly identifies Boston as having the largest concentration of peak-window players, consistent with their Eastern Conference Finals appearance. This framework provides actionable insight for front offices: roster construction around simultaneous peak-window concentration is measurable and correlates with real competitive outcomes.

1 Introduction

Professional basketball is a sport defined by windows. Teams rise, peak, and decline in cycles driven by the aging curves of their players. A franchise that misidentifies when its core is at its best risks squandering its competitive window through misaligned contracts, premature rebuilds, or ill-timed roster moves. Conversely, teams that correctly identify their window—and aggressively build around it—tend to be those that contend for championships.

Despite the richness of modern NBA statistical data, the question of *when* a player is in their peak performance window has received relatively little systematic

study. Most analyses treat player performance at a single point in time rather than as a trajectory, and few frameworks explicitly aggregate individual career arc predictions to generate team-level competitive window scores. This paper addresses these gaps with a unified machine learning pipeline. Our specific contributions are:

- A league-normalized **composite performance score** combining scoring, efficiency, and impact metrics into a single per-season signal, enabling fair comparison across eras and roles.
- **K-Means clustering** on player-level career trajectory features to discover four interpretable archetypes, validated via the Elbow Method and Silhouette Score [5].
- Training and comparison of **three supervised classifiers**—Logistic Regression, Random Forest [3], and XGBoost [4]—achieving AUC of 0.8348.
- A **championship window score** aggregating individual peak predictions to rank teams by concentration of simultaneously peaking players.

The real-world motivation is direct: front offices need tools that go beyond per-game averages to assess whether enough of a roster’s players share a peak window to make a championship run viable.

2 Related Work

Player Performance and Aging Curves. Vaci et al. [1] applied Bayesian latent variable modeling to longitudinal NBA data and found that players tend to peak around age 27–28, highlighting heterogeneity across players and selection bias. Our approach complements this by clustering on trajectory shapes rather than fitting a single population-level curve. Sarlis and Tjortjis [2] showed that efficiency-based statistics (PER, Box Plus/Minus) are stronger predictors of winning than raw counting stats, motivating our composite score design.

Machine Learning for NBA Prediction. Nguyen et al. [6] compared traditional ML and deep learning for predicting player performance and All-Star selection, finding that ensemble tree methods consistently outperform linear baselines and neural networks on structured tabular NBA data—motivating our use of Random Forest and XGBoost. Yao et al. [7] proposed a two-stage framework combining K-Means career-trend clustering with LSTM performance prediction, demonstrating the utility of combining unsupervised and supervised approaches for modeling player trajectories.

Clustering in Sports Analytics. Prior work has applied K-Means and hierarchical clustering to segment players into positional or stylistic archetypes [2]. Our work extends this by clustering on *career-level* trajectory features—slope, peak timing, longevity—rather than single-season statistics, enabling characterization of developmental patterns rather than snapshot roles.

Our Contribution. Unlike prior studies focusing on single-season performance or career totals in isolation, we explicitly model whether a given season is a peak relative to a player’s own career arc and aggregate these predictions to produce a team-level championship window score. To our knowledge, no prior work combines trajectory clustering, peak-season classification, and team-level competitive window scoring in a unified framework.

3 Methodology

3.1 Dataset

Our dataset contains 12,844 player-season observations spanning 27 NBA seasons from 1996–97 through 2022–23, covering 2,551 unique players [8]. Each row includes biometric features (height, weight, age), biographical features (college, country, draft round, draft number), and per-season performance statistics (points, rebounds, assists, net rating, offensive and defensive rebound percentage, usage percentage, true shooting percentage, and assist percentage).

3.2 Data Preprocessing

Draft Round Cleaning. The `draft_round` column contained valid values (1, 2, Undrafted) alongside 50 artifact entries (values 3–8), which were removed. Valid values were encoded as integers: Undrafted = 0, Round 1 = 1, Round 2 = 2.

Missing Values. The `college` column had 1,854 missing entries (primarily international players), filled with the label `International`. No other column contained missing values.

Net Rating Clipping. Players with very few games (1–5) produced extreme net rating values (up to ± 300). We clipped net rating to $[-30, 30]$, covering all realistic full-season values.

Traded Player Resolution. Four players appeared twice in the same season due to mid-season trades, resolved by a games-played weighted average across all numeric statistics.

Player-Level Split for Leakage Prevention. We split data by player rather than by row, ensuring the test set contains entirely unseen players and preventing the model from exploiting player identity to inflate test accuracy.

3.3 Feature Engineering

Career Year. Seasons were numbered sequentially per player (career year 1 = rookie season), capturing position within a player’s career arc.

Composite Performance Score. For season i with performance vector $\mathbf{x}_i = [\text{pts}_i, \text{net_rating}_i, \text{ts_pct}_i, \text{usg_pct}_i]$, we z-score each component across all

seasons and take the mean:

$$z_{i,k} = \frac{x_{i,k} - \mu_k}{\sigma_k}, \quad \text{score}_i = \frac{1}{4} \sum_k z_{i,k} \quad (1)$$

A value of $\text{score}_i > 0$ indicates an above-average season; $\text{score}_i < 0$ indicates below-average.

Is-Peak Target Variable. Peak seasons are those where the composite score meets or exceeds the player’s own 67th percentile:

$$\text{is_peak}_i = \mathbf{1}[\text{score}_i \geq Q_{0.67}(\{\text{score}_j : j \in \text{same player as } i\})] \quad (2)$$

This player-relative threshold ensures bench players’ best seasons are flagged as peaks even if their absolute performance is below league average.

Career Slope. We fit a linear regression of composite score over career year and record the slope $\hat{\beta}$, where $\hat{\beta} > 0$ indicates a player still improving and $\hat{\beta} < 0$ indicates decline.

3.4 K-Means Clustering

We aggregated each player’s career into a feature vector $\mathbf{p} = [\text{peak_age}, \text{career_length}, \overline{\text{score}}, \hat{\beta}, \text{peak_career_y}]$ standardized to zero mean and unit variance. We applied K-Means [11] with $k \in \{2, \dots, 10\}$ and selected $k = 4$ based on the Elbow Method and a local Silhouette Score [5] maximum at $k = 4$ (score = 0.185) relative to $k = 3$ (0.177) and $k = 5$ (0.181). The objective minimizes within-cluster sum of squared distances: $J = \sum_{c=1}^k \sum_{\mathbf{p} \in C_c} \|\mathbf{p} - \boldsymbol{\mu}_c\|^2$.

3.5 Supervised Classification

We trained three models to predict **is_peak** from 16 season-level features including age, career year, all per-game statistics, and composite score.

Logistic Regression (baseline): log-odds of a peak season as a linear combination of features. **Random Forest** [3]: ensemble of $T = 200$ trees on bootstrap samples with majority vote prediction. **XGBoost** [4]: sequentially trained gradient-boosted trees minimizing cross-entropy loss with tree-complexity regularization $\Omega(h_m) = \gamma T_m + \frac{1}{2} \lambda \|\mathbf{w}_m\|^2$.

4 Experiments & Results

4.1 Experimental Setup

After filtering to players with at least 3 seasons, our dataset contained 11,410 observations across 1,557 players, split 80/20 by player into training (1,246 players, 9,129 seasons) and test (311 players, 2,281 seasons) sets. Logistic Regression used StandardScaler-normalized features via scikit-learn [9]; tree-based models used raw features. All experiments used `random_state=42`.

4.2 Clustering Results

K-Means with $k = 4$ produced four interpretable archetypes. Figure 1 shows the PCA 2D projection [12] of the player feature space (PC1: 31.7% variance; PC2: 15.5%; combined: 47.2%). Table 1 summarizes each archetype.

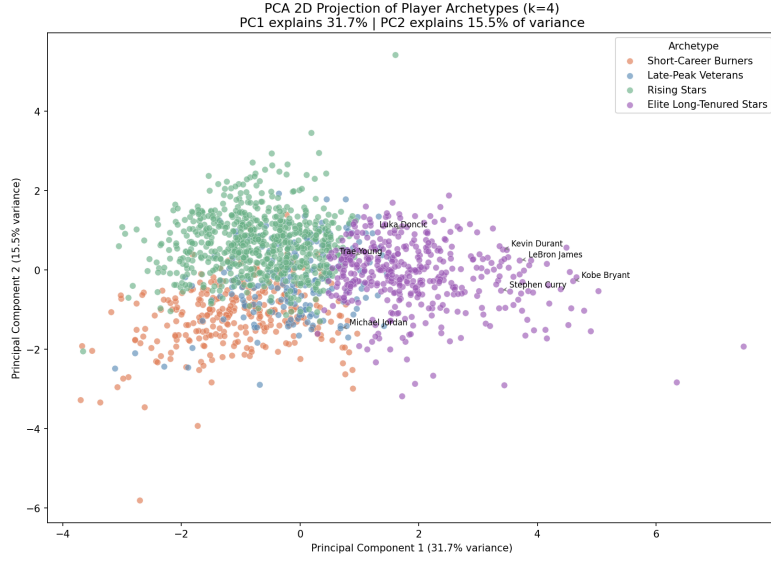


Figure 1: PCA 2D projection of player career trajectory features, colored by K-Means cluster assignment ($k = 4$). Elite Long-Tenured Stars (purple) separate clearly on PC1, reflecting their superior career-length and average performance scores.

Table 1: Summary of the four player trajectory archetypes discovered by K-Means.

Archetype	N	Avg Score	Peak Age	Career Len.	Slope
Short-Career Burners	284	-0.409	26.0	5.1	-0.124
Late-Peak Veterans	192	-0.202	31.3	5.9	-0.149
Rising Stars	662	-0.229	24.0	5.7	+0.008
Elite Long-Tenured Stars	419	+0.288	27.9	12.0	+0.009

Short-Career Burners (N=284) averaged only 5.1 seasons with negative career slopes (e.g., Kendrick Nunn, Jeremy Lin). **Late-Peak Veterans** (N=192) peaked after age 31 (e.g., Karl Malone at 33, Reggie Miller at 32). **Rising Stars** (N=662) peaked around age 24 with positive slopes indicating continued improvement (e.g., Luka Dončić, Ja Morant, Trae Young). **Elite Long-Tenured Stars** (N=419) show the highest average scores, longest careers (12 seasons

on average), and sustained performance (e.g., LeBron James, Kevin Durant, Stephen Curry).

Figure 2 shows the Elbow and Silhouette plots confirming the selection of $k = 4$.

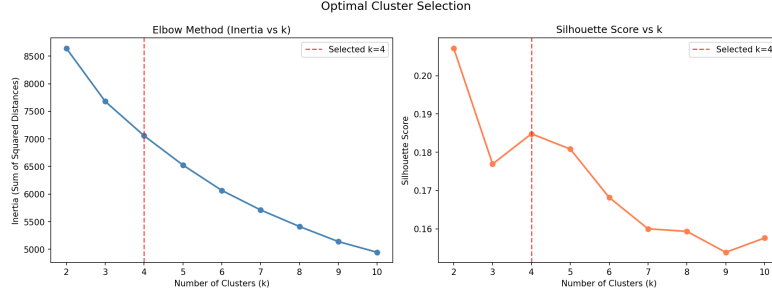


Figure 2: Elbow Method (left) and Silhouette Score (right) for $k = 2$ to 10. The red dashed line marks the selected $k = 4$.

4.3 Classification Results

Table 2 presents classifier performance on the held-out test set. XGBoost achieves the best AUC (0.835) and F1-Score (0.630) and is our selected model for championship window analysis. Random Forest matches closely, while Logistic Regression provides a useful linear baseline but exhibits the lowest Recall (0.509), missing nearly half of true peak seasons. ROC curves (Figure 3) confirm XGBoost maintains a consistent advantage across all false positive rate thresholds.

Table 2: Classification results for peak season prediction on the test set (311 unseen players).

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.753	0.663	0.509	0.576	0.818
Random Forest	0.763	0.657	0.589	0.621	0.831
XGBoost	0.762	0.647	0.614	0.630	0.835

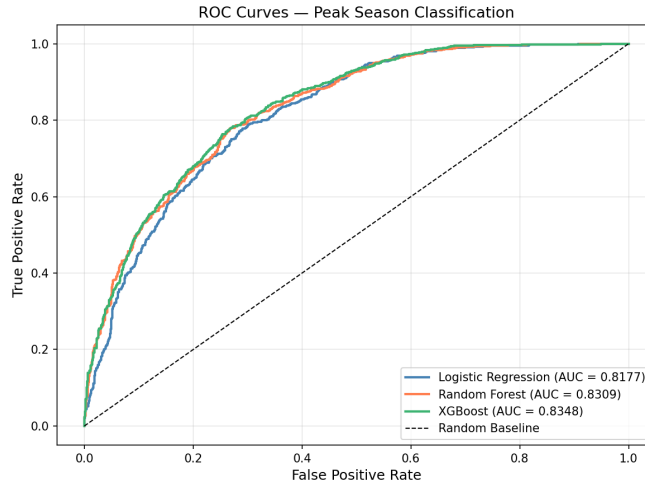


Figure 3: ROC curves for Logistic Regression, Random Forest, and XGBoost on the test set. XGBoost achieves the highest AUC of 0.8348.

4.4 Feature Importance Analysis

Figure 4 shows the top-10 feature importances from both models. Composite score dominates both (RF: 0.240; XGB: 0.286), confirming it is the strongest predictor of peak seasons. Net rating ranks second in Random Forest (0.143), while age and career year rank second and third in XGBoost (0.091 and 0.084). Both models agree that efficiency metrics outrank raw counting stats—peak seasons are defined by impact and efficiency, not volume. Draft round appears in XGBoost’s top six features (0.046), suggesting draft pedigree has a residual effect on peak-window attainment even years into a player’s career.

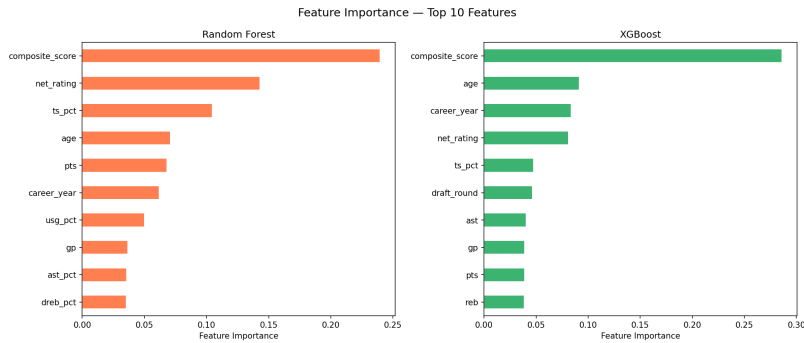


Figure 4: Top-10 feature importances from Random Forest (left) and XGBoost (right). Composite score dominates in both models; efficiency metrics outrank raw counting statistics.

4.5 Championship Window Analysis

We applied the XGBoost classifier to all 2022–23 season observations and summed, per team, the number of players predicted to be in their peak window (Figure 5). Boston leads with 10 simultaneously peaking players, followed by New York, Philadelphia, Phoenix, and Sacramento at 8 each. These predictions align with actual outcomes: Boston reached the Eastern Conference Finals; Philadelphia’s Joel Embiid won MVP; the Knicks and Celtics were recognized as legitimate Eastern Conference contenders; Phoenix and Sacramento both finished in the top four of their conference. The alignment supports the core hypothesis: peak-window concentration is a meaningful proxy for championship contention.

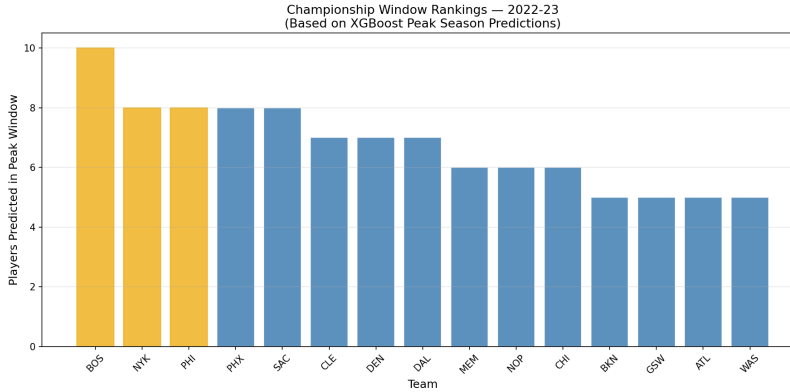


Figure 5: Championship Window Rankings for the 2022–23 NBA season. Boston (BOS) leads with 10 players simultaneously predicted in their peak window. Gold bars indicate the top 3 teams.

5 Conclusion

We presented a unified pipeline for NBA player trajectory modeling combining unsupervised archetype discovery with supervised peak-season classification and team-level competitive window scoring. K-Means identified four interpretable archetypes—Short-Career Burners, Late-Peak Veterans, Rising Stars, and Elite Long-Tenured Stars—characterizing distinct career developmental patterns across 27 seasons. XGBoost predicts peak seasons with $AUC = 0.835$, driven primarily by efficiency-based metrics over raw scoring volume. Applied to 2022–23 rosters, the championship window score correctly identifies Boston as the team with the largest concentration of simultaneously peaking players, consistent with their actual competitive outcomes.

Limitations. The dataset ends at 2022–23. The player-relative peak definition flags a below-average player’s best seasons even when their absolute output is low. The framework does not account for injury history, coaching quality, or

salary constraints—all of which influence competitive windows in practice.

Future Work. Natural extensions include time-series models (e.g., LSTM) to predict trajectory slope prospectively, SHAP [10] for per-prediction interpretability, and incorporating salary and contract data to identify teams whose championship window aligns with roster commitments.

The code and data for this project are available at: <https://github.com/pranavboddapati/NBACHampionshipWindowModel> and <https://github.com/vedhasvinod/nba-trajectory-modeling>

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